



**Generation Interconnection
System Impact Study Report
for
Queue Project AF1-132
SHINGLETOWN-BOALSBURG 46 KV
12 MW Capacity / 20 MW Energy**

August 2020

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is APS.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 General

The Interconnection Customer (IC), has proposed a Solar generating facility located in Centre County, Pennsylvania. The installed facilities will have a total capability of 20 MW with 12 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is April 01, 2021. This study does not imply a TO commitment to this in-service date.

Queue Number	AF1-132
Project Name	SHINGLETOWN-BOALSBURG 46 KV
State	Pennsylvania
County	Centre
Transmission Owner	APS
MFO	20
MWE	20
MWC	12
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

The AF1-132 project will interconnect with the West Penn Power (“WPP”) system. The interconnection of the project will be accomplished by tapping the Centre Hall – Shingletown 46 kV line and constructing a one span tap. The transmission line tap will be located approximately 6.5 miles from Shingletown substation. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct both the new interconnection line tap and the associated attachment facilities. The IC will also be responsible for the rough grade of the property and an access road to the proposed site. The project will also require non-direct connection upgrades at Center Hall and Shingletown substations.

Attachment 1 shows a one-line diagram of the proposed primary direct connection facilities for the AF1-132 generation project to connect to the FirstEnergy (“FE”) transmission system. IC will be responsible for constructing all of the facilities on its side of the POI, including the Attachment facilities which connect the generator to the FE transmission system.

5 Cost Summary

The AF1-132 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$307,975
Allocation towards System Network Upgrade Costs*	\$85,553,330 ¹
Total Costs	\$85,861,305

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation

¹ This project currently causes and contributes to overloads of the Transmission System (see Summer Peak Load Flow Analysis section below) and therefore has potential to have cost allocation for the system reinforcements listed in the report. The results may vary with queue customers withdrawing from the queue and other generators deactivating over time. If a customer is the first to cause the need for a project (causes loading to exceed 100% of rating), then the customer is responsible. If a customer’s MW contribution to a facility that is already overloaded by a prior queue is less than 5 MW, then they will likely not receive cost allocation (See PJM Manual 14A, Attachment B for Cost Allocation Rules).

contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

The AF1-132 project will interconnect with the West Penn Power (“WPP”) system. The interconnection of the project will be accomplished by tapping the Centre Hall – Shingletown 46 kV line and constructing a one span tap. The transmission line tap will be located approximately 6.5 miles from Shingletown substation. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct both the new interconnection line tap and the associated attachment facilities. The IC will also be responsible for the rough grade of the property and an access road to the proposed site. The project will also require non-direct connection upgrades at Center Hall and Shingletown substations.

The total physical interconnection costs is given in the table below:

Description	Total Cost
Tap the Center Hall – Shingletown 46 kV Line, install one span with a switch, and install 46 kV meter.	\$42,031
Upgrade relaying at Centre Hall Substation.	\$38,300
Upgrade relaying at Shingletown Substation.	\$38,300
Install two 46 kV network switches on the Centre Hall - Shingletown 46 kV Line	\$126,144
Review customer drawings, nameplates, and relay settings.	\$63,200
Total Physical Interconnection Costs	\$307,975

7 Schedule

Based on the scope of work for the interconnection facilities, it is expected to take a minimum of **24 months** after the signing of an Interconnection Agreement and construction kickoff call to complete the installation. This assumes that there will be no environmental issues with any of the new properties associated with this project, that there will be no delays in acquiring the necessary permits for implementing the defined work and that all system outages will be allowed when requested.

The schedule for the required Network Impact Reinforcements will be more clearly identified in future study phases. The estimated time to complete each of the required reinforcements is identified in the “System Reinforcements” section of the report.

8 Transmission Owner Analysis

FE performed an analysis of its underlying transmission <100 kV system. The AF1-132 project did not contribute to any overloads on the <100 kV FE Transmission System.

9 Interconnection Customer Requirements

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE’s “Requirements for Transmission Connected Facilities” document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>.

Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

The IC has requested a non-standard GSU transformer winding configuration. This transformer is in violation of section 14.2.6 of FE’s “Requirements for Transmission Connected Facilities” document and will not be accepted. The GSU transformer must have a delta or ungrounded wye connection on the high (utility) side, with the low side at the discretion of the IC.

9.1 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FE’s “Requirements for Transmission Connected Facilities” document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated 46 kV circuit breaker to protect the AF1-132 generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.

2. The purchase and installation of the minimum required FE generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the FE Transmission System Control Center.
4. Compliance with the FE and PJM generator power factor and voltage control requirements.
5. The execution of a back-up service agreement to serve the customer load supplied from the AF1-132 generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.

The IC will also be required to meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FE system.

9.2 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the FE transmission system.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Back Panel temperature (Fahrenheit)
- Irradiance (Watts/meter²)
- Ambient air temperature (Fahrenheit) – (Accepted, not required)
- Wind speed (meters/second) – (Accepted, not required)

Wind direction (decimal degrees from true north) – (Accepted, not required)

10.3 Interconnected Transmission Owner Requirements

The IC will be required to comply with all FE revenue metering requirements for generation interconnection customers which can be found in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>

11 Summer Peak Analysis

The Queue Project AF1-132 was evaluated as a 20.0 MW (Capacity 12.0 MW) injection tapping the Boalsburg to Shingleton 46 kV line in the APS area. Project AF1-132 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-132 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJE CT LOADIN G %	POST PROJE CT LOADIN G %	AC/D C	MW IMPA CT
41550528	200519	26REED TAP	115.0	PENEL EC	200522	26SHADE GP	115.0	PENEL EC	1	PL_P42_100548	breaker	225.0	99.21	100.26	AC	2.4
41550622	200714	26SHAW VL 1	115.0	PENEL EC	200713	26ROCKTON	115.0	PENEL EC	1	AP-P2-3-WP-230-446T	breaker	190.0	99.49	100.18	AC	1.55
41550623	200714	26SHAW VL 1	115.0	PENEL EC	200713	26ROCKTON	115.0	PENEL EC	1	AP-P2-2-WP-230-001T	breaker	190.0	99.4	100.09	AC	1.55
41807056	200714	26SHAW VL 1	115.0	PENEL EC	200713	26ROCKTON	115.0	PENEL EC	1	AP-P2-2-WP-230-001T	bus	190.0	99.4	100.09	AC	1.55
90786614	235175	01ELKO	230.0	AP	235971	01SQUABHL LW	230.0	AP	1	PN-P2-2-PN-230-007T	bus	621.0	81.22	82.4	AC	7.28
90786723	235175	01ELKO	230.0	AP	235971	01SQUABHL LW	230.0	AP	1	PN-P2-4-PN-230-14GT	breaker	621.0	81.45	82.63	AC	7.28
90786724	235175	01ELKO	230.0	AP	235971	01SQUABHL LW	230.0	AP	1	PN-P2-3-PN-230-14CT	breaker	621.0	81.24	82.43	AC	7.28
90786725	235175	01ELKO	230.0	AP	235971	01SQUABHL LW	230.0	AP	1	PN-P2-3-PN-230-14FT	breaker	621.0	81.24	82.43	AC	7.28
90786726	235175	01ELKO	230.0	AP	235971	01SQUABHL LW	230.0	AP	1	PN-P2-3-PN-230-14BT	breaker	621.0	81.22	82.4	AC	7.28
90786727	235175	01ELKO	230.0	AP	235971	01SQUABHL LW	230.0	AP	1	AP-P2-3-WP-230-460T	breaker	621.0	80.66	81.87	AC	7.4

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJE CT LOADIN G %	POST PROJE CT LOADIN G %	AC/D C	MW IMPA CT
41550630	200512	26LEWIST WN	115.0	PENEL EC	200519	26REED TAP	115.0	PENEL EC	1	PL_P42_100548	breaker	225.0	104.52	105.58	AC	2.4
41550316	200513	26LEWIST WN	230.0	PENEL EC	208005	JUNI BU2	230.0	PPL	1	PN-P2-3-PN-115-55A	breaker	624.0	114.93	116.01	AC	7.49
41806875	200513	26LEWIST WN	230.0	PENEL EC	208005	JUNI BU2	230.0	PPL	1	PN-P2-2-PN-115-032	bus	624.0	115.03	116.11	AC	7.49
41550311	200522	26SHADE GP	115.0	PENEL EC	938380	AE1-071 TAP	115.0	PENEL EC	1	PL_P42_100548	breaker	160.0	123.73	125.21	AC	2.4

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS T LOADING %	POST PROJECTIONS T LOADING %	AC/D C	MW IMPACT
412178 21	23524 8	01SHINGL	230.0	AP	20051 3	26LEWIST WN	230.0	PENEL EC	1	Base Case	single	491.0	117.33	118.51	AC	6.07
412178 23	23524 8	01SHINGL	230.0	AP	20051 3	26LEWIST WN	230.0	PENEL EC	1	PN-P1-2-PN-230-012A-A	single	570.0	112.28	113.36	AC	6.57
418067 93	93838 0	AE1-071 TAP	115.0	PENEL EC	20052 0	26ROXBURY	115.0	PENEL EC	1	PL_P22_100 582	bus	160.0	163.52	165.0	AC	2.4

11.4 Steady-State Voltage Requirements

To be determined

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS T LOADING %	POST PROJECTIONS T LOADING %	AC/D C	MW IMPACT
412184 25	2005 12	26LEWIST WN	115.0	PENEL EC	2005 19	26REED TAP	115.0	PENEL EC	1	PL_P12_100 618	operation	225.0	102.89	103.95	AC	2.4
412179 39	2005 13	26LEWIST WN	230.0	PENEL EC	2080 05	JUNI BU2	230.0	PPL	1	Base Case	operation	493.0	123.97	125.19	AC	6.82
412179 40	2005 13	26LEWIST WN	230.0	PENEL EC	2080 05	JUNI BU2	230.0	PPL	1	PN-P1-2-PN-115-094-A	operation	624.0	114.94	116.02	AC	7.5
412183 52	2005 19	26REED TAP	115.0	PENEL EC	2005 22	26SHADE GP	115.0	PENEL EC	1	PL_P12_100 618	operation	225.0	97.58	98.64	AC	2.4
412180 18	2005 22	26SHADE GP	115.0	PENEL EC	9383 80	AE1-071 TAP	115.0	PENEL EC	1	PL_P12_100 618	operation	160.0	121.49	122.97	AC	2.4
412181 11	2007 14	26SHAWV L 1	115.0	PENEL EC	2008 72	26BIOEN TP	115.0	PENEL EC	1	AP-P1-2-WP-230-323T	operation	185.0	105.39	106.5	AC	2.08
412181 62	2007 16	26PHILIPSB	115.0	PENEL EC	2009 04	26EAGL VAL	115.0	PENEL EC	1	AP-P1-2-WP-230-323T	operation	174.0	103.15	104.08	AC	1.93
412181 82	2007 27	26SHAW. 2	115.0	PENEL EC	2007 16	26PHILIPSB	115.0	PENEL EC	1	AP-P1-2-WP-230-323T	operation	185.0	103.56	104.44	AC	1.93
412179 71	2009 04	26EAGL VAL	115.0	PENEL EC	2005 27	26TYRONE N	115.0	PENEL EC	1	AP-P1-2-WP-230-323T	operation	191.0	136.11	136.96	AC	1.93
907869 20	2351 75	01ELKO	230.0	AP	2359 71	01SQUABHL LW	230.0	AP	1	AP-P1-3-WP-230-326T	operation	621.0	80.73	81.92	AC	7.28
412178 15	2352 48	01SHINGL	230.0	AP	2005 13	26LEWISTW N	230.0	PENEL EC	1	Base Case	operation	491.0	148.19	150.15	AC	10.11
412178 17	2352 48	01SHINGL	230.0	AP	2005 13	26LEWISTW N	230.0	PENEL EC	1	PN-P1-2-PN-230-012A-A	operation	570.0	145.26	147.13	AC	10.95
412180 22	9441 80	AF1-086 TAP	115.0	PENEL EC	2007 55	26WESTOVE R	115.0	PENEL EC	1	AP-P1-2-WP-230-323T	operation	237.0	99.31	100.16	AC	2.08

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-132	Upgrade Number
41550311	5	26SHADE GP 115.0 kV - AE1-071 TAP 115.0 kV Ckt 1	PENELEC: Upgrade no. 1: Reconductor 6.4 miles of line. Project Type : FAC Cost : \$19,564,800 New Ratings : 164.0/190.0/218.0 Time Estimate : 30 Months Queue Project AF1-132 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-132 could receive cost allocation. Note 2: Although Queue Project AF1-132 may not have cost responsibility for this upgrade, Queue Project AF1-132 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-132 comes into service prior to completion of the upgrade, Queue Project AF1-132 will need an interim study. Upgrade no. 2: Replace line trap at Shade Gap Project Type : FAC Cost : \$ 122,280 New Ratings : 176.0/218.0/248.0 Time Estimate : 12 Months Queue Project AF1-132 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-132 could receive cost allocation. Note 2: Although Queue Project AF1-132 may not have cost responsibility for this upgrade, Queue Project AF1-132 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-132 comes into service prior to completion of the upgrade, Queue Project AF1-112 will need an interim study.	\$19,687,080	\$0	n6821.1 & 6821.2

41217 821,41 21782 3	6	01SHINGL 230.0 kV - 26LEWISTWN 230.0 kV Ckt 1	PENELEC: Upgrade no. 1: Reconductor 26.12 miles of transmission line Project Type : FAC Cost : \$ 111,788,376 New Ratings : 548.0/688.0/791.0 Time Estimate : 48 Months The cost allocation table is below: <table border="1"> <thead> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> </thead> <tbody> <tr> <td>AF1-068</td><td>0.5</td><td>0.51%</td><td>\$ 567,579</td><td>single</td><td>SP</td><td>Base Case</td></tr> <tr> <td>AF1-099</td><td>29.6</td><td>32.68%</td><td>\$ 36,534,810</td><td>single</td><td>SP</td><td>Base Case</td></tr> <tr> <td>AF1-100</td><td>48.1</td><td>53.09%</td><td>\$ 59,349,016</td><td>single</td><td>SP</td><td>Base Case</td></tr> <tr> <td>AF1-112</td><td>6.4</td><td>7.02%</td><td>\$ 7,847,396</td><td>single</td><td>SP</td><td>Base Case</td></tr> <tr> <td>AF1-132</td><td>6.1</td><td>6.70%</td><td>\$ 7,489,574</td><td>single</td><td>SP</td><td>Base Case</td></tr> </tbody> </table> Upgrade no. 2: Adjust/replace line relaying at Lewistown Project Type : FAC Cost : \$ 489,120 New Ratings : 664/800/920 Time Estimate : 12 Months AF1-132 is the driver for this upgrade. APS: Upgrade no. 1: Replace 1033.5 ACSR & 2.00 IPS AL bus conductor at Shingletown substation. Project Type : FAC Cost : \$ 162,815 New Ratings : 546.0/666.0/666.0 Time Estimate : 48 Months	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AF1-068	0.5	0.51%	\$ 567,579	single	SP	Base Case	AF1-099	29.6	32.68%	\$ 36,534,810	single	SP	Base Case	AF1-100	48.1	53.09%	\$ 59,349,016	single	SP	Base Case	AF1-112	6.4	7.02%	\$ 7,847,396	single	SP	Base Case	AF1-132	6.1	6.70%	\$ 7,489,574	single	SP	Base Case	\$112,993,883	\$8,045,114	N6822.1 & N6822.2 & N6823.1 & N6823.2 & N6823.3
Queue	MW	Cost %	Cost \$	Type	Study	ContName																																										
AF1-068	0.5	0.51%	\$ 567,579	single	SP	Base Case																																										
AF1-099	29.6	32.68%	\$ 36,534,810	single	SP	Base Case																																										
AF1-100	48.1	53.09%	\$ 59,349,016	single	SP	Base Case																																										
AF1-112	6.4	7.02%	\$ 7,847,396	single	SP	Base Case																																										
AF1-132	6.1	6.70%	\$ 7,489,574	single	SP	Base Case																																										

The cost allocation table is below:

Queue	MW	Cost %	Cost \$	Type	Study	ContName
AF1-068	0.5	0.51%	\$ 827	single	SP	Base Case
AF1-099	29.6	32.68%	\$ 53,211	single	SP	Base Case
AF1-100	48.1	53.09%	\$ 86,439	single	SP	Base Case
AF1-112	6.4	7.02%	\$ 11,429	single	SP	Base Case
AF1-132	6.1	6.70%	\$ 10,908	single	SP	Base Case

Upgrade no. 2:

Replace risers at Shingletown substation.

Project Type : FAC

Cost : \$ 162,815

New Ratings : 548.0/688.0/791.0

Time Estimate : 12 Months

The cost allocation table is below:

Queue	MW	Cost %	Cost \$	Type	Study	ContName
AF1-100	48.1	79.46%	\$ 129,380	single	SP	Base Case
AF1-112	6.4	10.51%	\$ 17,107	single	SP	Base Case
AF1-132	6.1	10.03%	\$ 16,327	single	SP	Base Case

Upgrade no. 3:

Replace 1200 A line side and bus side disconnect switches at Shingletown substation. Adjust line relaying as necessary at Shingletown substation.

Project Type : FAC

Cost : \$ 390,757

New Ratings : 664.0/800.0/800.0

Time Estimate : 12 Months

The cost allocation table is below:

Queue	MW	Cost %	Cost \$	Type	Study	ContName

ID	Idx	Facility	Upgrade Description							Cost	Cost Allocated to AF1-132	Upgrade Number
			AF1-100	48.1	79.46%	\$ 310,514	single	SP	Base Case			
			AF1-112	6.36	10.51%	\$ 41,058	single	SP	Base Case			
			AF1-132	6.07	0.10	\$ 39,185	single	SP	Base Case			
41807 056,41 55062 3,4155 0622	2	26SHAWVL 1 115.0 kV - 26ROCKTON 115.0 kV Ckt 1	PENELEC: Replace line trap at Shawville. Project Type : FAC Cost : \$ 122,280 New Ratings : 167.0/202.0/227.0 Time Estimate : 12 months AF1-132 is the driver for this upgrade.							\$122,280	\$122,280	N6847

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-132	Upgrade Number
41550630	3	26LEWISTWN 115.0 kV - 26REED TAP 115.0 kV Ckt 1	PENELEC: Rebuild 11.84 miles transmission line Project Type : FAC Cost : \$36,194,880 New Ratings: 202.0/245.0/228.0 Time Estimate : 36 Months Queue Project AF1-132 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-132 could receive cost allocation. Note 2: Although Queue Project AF1-132 may not have cost responsibility for this upgrade, Queue Project AF1-132 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-132 comes into service prior to completion of the upgrade, Queue Project AF1-132 will need an interim study.	\$36,194,880	\$0	N6812.1

41806 793	7	AE1-071 TAP 115.0 kV - 26ROXBURY 115.0 kV Ckt 1	PENELEC: Upgrade no. 1: Reconductor 6.4 miles of line. Project Type : FAC Cost : \$15,651,840 New Ratings : 147/191/214 Time Estimate : 36 Months Queue Project AF1-132 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-132 could receive cost allocation. Note 2: Although Queue Project AF1-132 may not have cost responsibility for this upgrade, Queue Project AF1-132 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-132 comes into service prior to completion of the upgrade, Queue Project AF1-132 will need an interim study. Upgrade no. 2: Replace substation conductor at Roxbury Project Type : FAC Cost : \$91,710 New Ratings : 176/218/248 Time Estimate : 12 Months Queue Project AF1-132 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-132 could receive cost allocation. Note 2: Although Queue Project AF1-132 may not have cost responsibility for this upgrade, Queue Project AF1-132 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-132 comes into service prior to completion of the upgrade, Queue Project AF1-132 will need an interim study. Upgrade no. 3: Replace transmission line drop at Roxbury Project Type : FAC	\$16,568,940	\$0	n6493 & n6493. 1 & n6493. 2 & n6493. 3
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ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-132	Upgrade Number
			Cost : \$91,710 New Ratings : 176/224/258 Time Estimate : 12 Months Queue Project AF1-132 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-132 could receive cost allocation. Note 2: Although Queue Project AF1-132 may not have cost responsibility for this upgrade, Queue Project AF1-132 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-132 comes into service prior to completion of the upgrade, Queue Project AF1-132 will need an interim study. Upgrade no. 4: Replace circuit breaker and CTs at Roxbury Project Type : FAC Cost : \$733,680 New Ratings : 432/442/457 Time Estimate : 12 Months Queue Project AF1-132 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-132 could receive cost allocation. Note 2: Although Queue Project AF1-132 may not have cost responsibility for this upgrade, Queue Project AF1-132 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-132 comes into service prior to completion of the upgrade, Queue Project AF1-132 will need an interim study.			

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-132	Upgrade Number																																																																						
41550 316,41 80687 5	4	26LEWISTWN 230.0 kV - JUNI BU2 230.0 kV Ckt 1	PENELEC: Rebuild 24.59 miles of transmission line Project Type : FAC Cost : \$105,240,282 New Ratings : 815.0/829.0/858.0 Time Estimate : 48 Months The cost allocation table is below: <table border="1"> <thead> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> </thead> <tbody> <tr> <td>AF1-099</td><td>31.6</td><td>28.75%</td><td>\$ 30,256,222</td><td>bus</td><td>SP</td><td>PN-P2-2-PN-115-032</td></tr> <tr> <td>AF1-100</td><td>63.4</td><td>57.71%</td><td>\$ 60,732,872</td><td>bus</td><td>SP</td><td>PN-P2-2-PN-115-032</td></tr> <tr> <td>AF1-112</td><td>7.4</td><td>6.72%</td><td>\$ 7,072,883</td><td>bus</td><td>SP</td><td>PN-P2-2-PN-115-032</td></tr> <tr> <td>AF1-132</td><td>7.5</td><td>6.82%</td><td>\$ 7,178,305</td><td>bus</td><td>SP</td><td>PN-P2-2-PN-115-032</td></tr> </tbody> </table> PPL: Rebuild PPL's ~0.9mi LEWI-JUNI 230kV Line and replace terminal equipment at Juniata 230kV Project Type : FAC Cost : \$7,000,000 New Ratings : 1245/1431/1431 Time Estimate : 24 Months The cost allocation table is below: <table border="1"> <thead> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> </thead> <tbody> <tr> <td>AF1-099</td><td>31.6</td><td>28.75%</td><td>\$ 2,012,476</td><td>bus</td><td>SP</td><td>PN-P2-2-PN-115-032</td></tr> <tr> <td>AF1-100</td><td>63.4</td><td>57.71%</td><td>\$ 4,039,614</td><td>bus</td><td>SP</td><td>PN-P2-2-PN-115-032</td></tr> <tr> <td>AF1-112</td><td>7.4</td><td>6.72%</td><td>\$ 470,449</td><td>bus</td><td>SP</td><td>PN-P2-2-PN-115-032</td></tr> <tr> <td>AF1-132</td><td>7.5</td><td>6.82%</td><td>\$ 477,461</td><td>bus</td><td>SP</td><td>PN-P2-2-PN-115-032</td></tr> </tbody> </table>	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AF1-099	31.6	28.75%	\$ 30,256,222	bus	SP	PN-P2-2-PN-115-032	AF1-100	63.4	57.71%	\$ 60,732,872	bus	SP	PN-P2-2-PN-115-032	AF1-112	7.4	6.72%	\$ 7,072,883	bus	SP	PN-P2-2-PN-115-032	AF1-132	7.5	6.82%	\$ 7,178,305	bus	SP	PN-P2-2-PN-115-032	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AF1-099	31.6	28.75%	\$ 2,012,476	bus	SP	PN-P2-2-PN-115-032	AF1-100	63.4	57.71%	\$ 4,039,614	bus	SP	PN-P2-2-PN-115-032	AF1-112	7.4	6.72%	\$ 470,449	bus	SP	PN-P2-2-PN-115-032	AF1-132	7.5	6.82%	\$ 477,461	bus	SP	PN-P2-2-PN-115-032	\$112,240,282	\$7,655,766	N6817.1 & N6818
Queue	MW	Cost %	Cost \$	Type	Study	ContName																																																																						
AF1-099	31.6	28.75%	\$ 30,256,222	bus	SP	PN-P2-2-PN-115-032																																																																						
AF1-100	63.4	57.71%	\$ 60,732,872	bus	SP	PN-P2-2-PN-115-032																																																																						
AF1-112	7.4	6.72%	\$ 7,072,883	bus	SP	PN-P2-2-PN-115-032																																																																						
AF1-132	7.5	6.82%	\$ 7,178,305	bus	SP	PN-P2-2-PN-115-032																																																																						
Queue	MW	Cost %	Cost \$	Type	Study	ContName																																																																						
AF1-099	31.6	28.75%	\$ 2,012,476	bus	SP	PN-P2-2-PN-115-032																																																																						
AF1-100	63.4	57.71%	\$ 4,039,614	bus	SP	PN-P2-2-PN-115-032																																																																						
AF1-112	7.4	6.72%	\$ 470,449	bus	SP	PN-P2-2-PN-115-032																																																																						
AF1-132	7.5	6.82%	\$ 477,461	bus	SP	PN-P2-2-PN-115-032																																																																						

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-132	Upgrade Number
41550 528	1	26REED TAP 115.0 kV - 26SHADE GP 115.0 kV Ckt 1	PENELEC: Upgrade no. 1: Rebuild 22.36 miles of transmission line. Project Type : FAC Cost : \$67,957,110 New Ratings: 193.0/240.0/269.0 Time Estimate : 42 Months AF1-132 is the driver for this upgrade.	\$69,730,170	\$69,730,170	N6820.1
			TOTAL COST	\$367,537,515	\$85,553,330	

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41550528	200519	26REED TAP	PENELEC	200522	26SHADE GP	PENELEC	1	PL_P42_100548	breaker	225.0	99.21	100.26	AC	2.4

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200812	26ALY HYDR	0.4762	50/50	0.4762
200852	26WARR RDG	0.1361	50/50	0.1361
235003	AC1-025 E	0.1804	50/50	0.1804
236828	01GRAYMONT	0.4640	50/50	0.4640
290086	Q-036 E	2.0495	Adder	2.41
293301	N-039 E	3.5676	Adder	4.2
294515	O38_P22	3.1216	Adder	3.67
919491	AA2-000	26.1120	Adder	30.72
930511	AB1-092	0.9588	Adder	1.13
936421	AD2-055	1.9739	Adder	2.32
936991	AD2-133 C	0.9223	Adder	1.09
936992	AD2-133 E	4.2184	Adder	4.96
939171	AE1-147 C	1.3841	50/50	1.3841
939172	AE1-147 E	0.9227	50/50	0.9227
940201	AE2-001 C	1.3886	50/50	1.3886
940202	AE2-001 E	0.9258	50/50	0.9258
940671	AE2-054 C	-0.5515	Adder	-0.65
940681	AE2-055 C	1.4580	50/50	1.4580
940682	AE2-055 E	0.9720	50/50	0.9720
941231	AE2-117 C	1.5364	50/50	1.5364
941232	AE2-117 E	1.0242	50/50	1.0242
941241	AE2-118 C	1.5364	50/50	1.5364
941242	AE2-118 E	1.0242	50/50	1.0242
941261	AE2-120 C	1.3910	50/50	1.3910
941262	AE2-120 E	0.9274	50/50	0.9274
941271	AE2-121 C	0.7317	50/50	0.7317
941272	AE2-121 E	0.4885	50/50	0.4885
941321	AE2-126 C	0.6617	Adder	0.78
941322	AE2-126 E	0.4411	Adder	0.52
941331	AE2-129 C	0.6652	Adder	0.78
941332	AE2-129 E	0.4435	Adder	0.52
941351	AE2-131 C	0.6652	Adder	0.78
941352	AE2-131 E	0.4435	Adder	0.52
942351	AE2-248 C	1.1433	50/50	1.1433
942352	AE2-248 E	0.7622	50/50	0.7622
942491	AE2-262 C	4.0855	Adder	4.81
942492	AE2-262 E	2.7455	Adder	3.23
942501	AE2-263 C	3.8404	Adder	4.52
942502	AE2-263 E	2.5641	Adder	3.02
942511	AE2-264 C	5.0510	50/50	5.0510
942512	AE2-264 E	3.3674	50/50	3.3674
943751	AF1-043	5.9217	Adder	6.97

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
944001	AF1-068 C O1	1.5337	50/50	1.5337
944002	AF1-068 E O1	0.8627	50/50	0.8627
944311	AF1-099 C	6.2018	Adder	7.3
944312	AF1-099 E	4.1346	Adder	4.86
944321	AF1-100 C	10.4754	Adder	12.32
944322	AF1-100 E	6.9836	Adder	8.22
944471	AF1-112 C	1.5163	50/50	1.5163
944472	AF1-112 E	0.8529	50/50	0.8529
944671	AF1-132 C O1	1.4428	50/50	1.4428
944672	AF1-132 E O1	0.9618	50/50	0.9618
944691	AF1-134 C O1	0.5643	Adder	0.66
944692	AF1-134 E O1	0.3762	Adder	0.44
944771	AF1-142 C	9.9229	Adder	11.67
944772	AF1-142 E	6.6153	Adder	7.78
944841	AF1-149 C	1.3914	50/50	1.3914
944842	AF1-149 E	0.9276	50/50	0.9276
944881	AF1-153 C O1	0.6331	Adder	0.74
944882	AF1-153 E O1	0.4221	Adder	0.5
944901	AF1-155 C	0.6293	Adder	0.74
944902	AF1-155 E	0.4196	Adder	0.49
945161	AF1-181	0.0561	Adder	0.07
945171	AF1-182	0.2810	Adder	0.33
945181	AF1-183	0.0655	Adder	0.08
945491	AF1-214 C	1.3909	50/50	1.3909
945492	AF1-214 E	0.9273	50/50	0.9273
945551	AF1-220 C	4.6797	Adder	5.51
945552	AF1-220 E	3.1215	Adder	3.67
946421	AF1-306 C	2.7481	Adder	3.23
946422	AF1-306 E	10.9925	Adder	12.93
WEC	WEC	0.0400	Confirmed LTF	0.0400
LGEE	LGEE	0.0593	Confirmed LTF	0.0593
CBM-W2	CBM-W2	0.4996	Confirmed LTF	0.4996
TVA	TVA	0.0238	Confirmed LTF	0.0238
O-066	O-066	1.5926	Confirmed LTF	1.5926
CHEOAH	CHEOAH	0.0110	Confirmed LTF	0.0110
CBM-S1	CBM-S1	0.2812	Confirmed LTF	0.2812
G-007	G-007	0.2829	Confirmed LTF	0.2829
MEC	MEC	0.1621	Confirmed LTF	0.1621
CALDERWOOD	CALDERWOOD	0.0094	Confirmed LTF	0.0094
CATAWBA	CATAWBA	0.0427	Confirmed LTF	0.0427
CBM-W1	CBM-W1	1.9391	Confirmed LTF	1.9391

11.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41807056	200714	26SHAWVL 1	PENELEC	200713	26ROCKTON	PENELEC	1	AP-P2-2-WP-230-001T	bus	190.0	99.4	100.09	AC	1.55

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200665	26SHAWVL 3	3.4762	50/50	3.4762
200666	26SHAWVL 4	3.4605	50/50	3.4605
200715	26SHAWVL 1	3.1498	50/50	3.1498
200722	26SHAWVL 2	3.2303	50/50	3.2303
200905	26Q36	0.1903	50/50	0.1903
200913	26SHAW-D	0.1917	50/50	0.1917
235003	AC1-025 E	0.0988	Adder	0.12
236828	01GRAYMONT	0.2708	Adder	0.32
290086	Q-036 E	4.2972	50/50	4.2972
919491	AA2-000	28.7055	Adder	33.77
930511	AB1-092	1.0540	Adder	1.24
936421	AD2-055	2.1700	Adder	2.55
936991	AD2-133 C	1.9337	50/50	1.9337
936992	AD2-133 E	8.8447	50/50	8.8447
939171	AE1-147 C	0.8160	Adder	0.96
939172	AE1-147 E	0.5440	Adder	0.64
940201	AE2-001 C	0.8141	Adder	0.96
940202	AE2-001 E	0.5427	Adder	0.64
940681	AE2-055 C	0.7836	Adder	0.92
940682	AE2-055 E	0.5224	Adder	0.61
941261	AE2-120 C	0.8129	Adder	0.96
941262	AE2-120 E	0.5420	Adder	0.64
941271	AE2-121 C	0.4357	Adder	0.51
941272	AE2-121 E	0.2909	Adder	0.34
941331	AE2-129 C	1.6774	50/50	1.6774
941332	AE2-129 E	1.1182	50/50	1.1182
941351	AE2-131 C	1.6774	50/50	1.6774
941352	AE2-131 E	1.1182	50/50	1.1182
942351	AE2-248 C	0.6371	Adder	0.75
942352	AE2-248 E	0.4247	Adder	0.5
942491	AE2-262 C	3.8203	Adder	4.49
942492	AE2-262 E	2.5673	Adder	3.02
942501	AE2-263 C	3.5911	Adder	4.22
942502	AE2-263 E	2.3976	Adder	2.82
943751	AF1-043	6.5099	Adder	7.66
944001	AF1-068 C O1	0.8453	Adder	0.99
944002	AF1-068 E O1	0.4755	Adder	0.56
944181	AF1-086 C O1	1.1792	Adder	1.39
944182	AF1-086 E O1	5.1301	Adder	6.04
944311	AF1-099 C	5.7993	Adder	6.82
944312	AF1-099 E	3.8662	Adder	4.55

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
944321	AF1-100 C	16.4940	50/50	16.4940
944322	AF1-100 E	10.9960	50/50	10.9960
944382	AF1-103 BAT	1.3516	Merchant Transmission	1.3516
944471	AF1-112 C	0.8529	Adder	1.0
944472	AF1-112 E	0.4797	Adder	0.56
944671	AF1-132 C O1	0.7903	Adder	0.93
944672	AF1-132 E O1	0.5269	Adder	0.62
944691	AF1-134 C O1	1.3916	50/50	1.3916
944692	AF1-134 E O1	0.9278	50/50	0.9278
944771	AF1-142 C	9.2788	Adder	10.92
944772	AF1-142 E	6.1859	Adder	7.28
944841	AF1-149 C	0.8128	Adder	0.96
944842	AF1-149 E	0.5419	Adder	0.64
945161	AF1-181	0.0913	50/50	0.0913
945171	AF1-182	0.4570	50/50	0.4570
945181	AF1-183	0.1458	50/50	0.1458
945491	AF1-214 C	0.8129	Adder	0.96
945492	AF1-214 E	0.5420	Adder	0.64
946423	AF1-306 BAT	32.2299	50/50	32.2299
NEWTON	NEWTON	0.2869	Confirmed LTF	0.2869
FARMERCITY	FARMERCITY	0.0147	Confirmed LTF	0.0147
G-007A	G-007A	1.1364	Confirmed LTF	1.1364
VFT	VFT	3.0831	Confirmed LTF	3.0831
NY	NY	0.0348	Confirmed LTF	0.0348
PRAIRIE	PRAIRIE	0.6664	Confirmed LTF	0.6664
COFFEEN	COFFEEN	0.0535	Confirmed LTF	0.0535
CHEOAH	CHEOAH	0.1016	Confirmed LTF	0.1016
EDWARDS	EDWARDS	0.0977	Confirmed LTF	0.0977
TILTON	TILTON	0.1739	Confirmed LTF	0.1739
MADISON	MADISON	0.0202	Confirmed LTF	0.0202
GIBSON	GIBSON	0.1469	Confirmed LTF	0.1469
CALDERWOOD	CALDERWOOD	0.1019	Confirmed LTF	0.1019
BLUEG	BLUEG	0.4618	Confirmed LTF	0.4618
TRIMBLE	TRIMBLE	0.1486	Confirmed LTF	0.1486
CATAWBA	CATAWBA	0.0549	Confirmed LTF	0.0549

11.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41550630	200512	26LEWISTWN	PENELEC	200519	26REED TAP	PENELEC	1	PL_P42_100548	breaker	225.0	104.52	105.58	AC	2.4

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200812	26ALY HYDR	0.4762	50/50	0.4762
200852	26WARR RDG	0.1361	50/50	0.1361
235003	AC1-025 E	0.1804	50/50	0.1804
236828	01GRAYMONT	0.4640	50/50	0.4640
290086	Q-036 E	2.0495	Adder	2.41
293301	N-039 E	3.5676	Adder	4.2
294515	O38_P22	3.1216	Adder	3.67
919491	AA2-000	26.1120	Adder	30.72
930511	AB1-092	0.9588	Adder	1.13
936421	AD2-055	1.9739	Adder	2.32
936991	AD2-133 C	0.9223	Adder	1.09
936992	AD2-133 E	4.2184	Adder	4.96
939171	AE1-147 C	1.3841	50/50	1.3841
939172	AE1-147 E	0.9227	50/50	0.9227
940201	AE2-001 C	1.3886	50/50	1.3886
940202	AE2-001 E	0.9258	50/50	0.9258
940671	AE2-054 C	-0.5515	Adder	-0.65
940681	AE2-055 C	1.4580	50/50	1.4580
940682	AE2-055 E	0.9720	50/50	0.9720
941231	AE2-117 C	1.5364	50/50	1.5364
941232	AE2-117 E	1.0242	50/50	1.0242
941241	AE2-118 C	1.5364	50/50	1.5364
941242	AE2-118 E	1.0242	50/50	1.0242
941261	AE2-120 C	1.3910	50/50	1.3910
941262	AE2-120 E	0.9274	50/50	0.9274
941271	AE2-121 C	0.7317	50/50	0.7317
941272	AE2-121 E	0.4885	50/50	0.4885
941321	AE2-126 C	0.6617	Adder	0.78
941322	AE2-126 E	0.4411	Adder	0.52
941331	AE2-129 C	0.6652	Adder	0.78
941332	AE2-129 E	0.4435	Adder	0.52
941351	AE2-131 C	0.6652	Adder	0.78
941352	AE2-131 E	0.4435	Adder	0.52
942351	AE2-248 C	1.1433	50/50	1.1433
942352	AE2-248 E	0.7622	50/50	0.7622
942491	AE2-262 C	4.0855	Adder	4.81
942492	AE2-262 E	2.7455	Adder	3.23
942501	AE2-263 C	3.8404	Adder	4.52
942502	AE2-263 E	2.5641	Adder	3.02
942511	AE2-264 C	5.0510	50/50	5.0510
942512	AE2-264 E	3.3674	50/50	3.3674
943751	AF1-043	5.9217	Adder	6.97

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
944001	AF1-068 C O1	1.5337	50/50	1.5337
944002	AF1-068 E O1	0.8627	50/50	0.8627
944311	AF1-099 C	6.2018	Adder	7.3
944312	AF1-099 E	4.1346	Adder	4.86
944321	AF1-100 C	10.4754	Adder	12.32
944322	AF1-100 E	6.9836	Adder	8.22
944471	AF1-112 C	1.5163	50/50	1.5163
944472	AF1-112 E	0.8529	50/50	0.8529
944671	AF1-132 C O1	1.4428	50/50	1.4428
944672	AF1-132 E O1	0.9618	50/50	0.9618
944691	AF1-134 C O1	0.5643	Adder	0.66
944692	AF1-134 E O1	0.3762	Adder	0.44
944771	AF1-142 C	9.9229	Adder	11.67
944772	AF1-142 E	6.6153	Adder	7.78
944841	AF1-149 C	1.3914	50/50	1.3914
944842	AF1-149 E	0.9276	50/50	0.9276
944881	AF1-153 C O1	0.6331	Adder	0.74
944882	AF1-153 E O1	0.4221	Adder	0.5
944901	AF1-155 C	0.6293	Adder	0.74
944902	AF1-155 E	0.4196	Adder	0.49
945161	AF1-181	0.0561	Adder	0.07
945171	AF1-182	0.2810	Adder	0.33
945181	AF1-183	0.0655	Adder	0.08
945491	AF1-214 C	1.3909	50/50	1.3909
945492	AF1-214 E	0.9273	50/50	0.9273
945551	AF1-220 C	4.6797	Adder	5.51
945552	AF1-220 E	3.1215	Adder	3.67
946421	AF1-306 C	2.7481	Adder	3.23
946422	AF1-306 E	10.9925	Adder	12.93
WEC	WEC	0.0400	Confirmed LTF	0.0400
LGEE	LGEE	0.0593	Confirmed LTF	0.0593
CBM-W2	CBM-W2	0.4996	Confirmed LTF	0.4996
TVA	TVA	0.0238	Confirmed LTF	0.0238
O-066	O-066	1.5926	Confirmed LTF	1.5926
CHEOAH	CHEOAH	0.0110	Confirmed LTF	0.0110
CBM-S1	CBM-S1	0.2812	Confirmed LTF	0.2812
G-007	G-007	0.2829	Confirmed LTF	0.2829
MEC	MEC	0.1621	Confirmed LTF	0.1621
CALDERWOOD	CALDERWOOD	0.0094	Confirmed LTF	0.0094
CATAWBA	CATAWBA	0.0427	Confirmed LTF	0.0427
CBM-W1	CBM-W1	1.9391	Confirmed LTF	1.9391

11.7.4 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41806875	200513	26LEWISTWN	PENELEC	208005	JUNI BU2	PPL	1	PN-P2-2-PN-115-032	bus	624.0	115.03	116.11	AC	7.49

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200649	26PENNTech	1.3657	50/50	1.3657
200665	26SHAWVL 3	7.7517	50/50	7.7517
200666	26SHAWVL 4	7.7260	50/50	7.7260
200715	26SHAWVL 1	5.1349	50/50	5.1349
200722	26SHAWVL 2	5.2664	50/50	5.2664
200812	26ALY HYDR	1.4884	50/50	1.4884
200852	26WARR RDG	0.4253	50/50	0.4253
200883	Q-053 E	6.9106	50/50	6.9106
200886	26ARWF_N39	0.5723	50/50	0.5723
200894	26K02	7.5298	50/50	7.5298
200898	26AA1-106	1.7248	Adder	2.03
200905	26Q36	0.3279	50/50	0.3279
200913	26SHAW-D	0.2330	50/50	0.2330
200915	26CHSTN_FL	0.1938	50/50	0.1938
201477	26Y2-055	2.6671	Adder	3.14
203034	26NA_O38_P22	0.5008	50/50	0.5008
235003	AC1-025 E	0.5621	50/50	0.5621
236828	01GRAYMONT	1.4447	50/50	1.4447
290086	Q-036 E	7.4056	50/50	7.4056
293301	N-039 E	12.9261	50/50	12.9261
293393	V3-030E	2.7911	Adder	3.28
293603	O-018 E	4.7228	Adder	5.56
294515	O38_P22	11.3103	50/50	11.3103
296332	R-032 E	5.4494	Adder	6.41
919491	AA2-000	92.8732	50/50	92.8732
922932	AB1-082 OP (Suspended)	1.7668	Adder	2.08
930511	AB1-092	3.4100	50/50	3.4100
936421	AD2-055	7.0206	50/50	7.0206
936991	AD2-133 C	3.3325	50/50	3.3325
936992	AD2-133 E	15.2426	50/50	15.2426
938381	AE1-071 C	34.6729	50/50	34.6729
938382	AE1-071 E	21.2169	50/50	21.2169
938991	AE1-128 C	6.3281	Adder	7.44
938992	AE1-128 E	4.2187	Adder	4.96
939171	AE1-147 C	4.3085	50/50	4.3085
939172	AE1-147 E	2.8723	50/50	2.8723
940201	AE2-001 C	4.3230	50/50	4.3230
940202	AE2-001 E	2.8820	50/50	2.8820
940681	AE2-055 C	4.5445	50/50	4.5445
940682	AE2-055 E	3.0297	50/50	3.0297

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
940861	AE2-074 C	1.3415	Adder	1.58
940862	AE2-074 E	1.7659	Adder	2.08
941191	AE2-113 C	9.0369	50/50	9.0369
941192	AE2-113 E	9.7299	50/50	9.7299
941231	AE2-117 C	4.8020	50/50	4.8020
941232	AE2-117 E	3.2014	50/50	3.2014
941241	AE2-118 C	4.8020	50/50	4.8020
941242	AE2-118 E	3.2014	50/50	3.2014
941261	AE2-120 C	4.3309	50/50	4.3309
941262	AE2-120 E	2.8873	50/50	2.8873
941271	AE2-121 C	2.2772	50/50	2.2772
941272	AE2-121 E	1.5205	50/50	1.5205
941321	AE2-126 C	2.3890	50/50	2.3890
941322	AE2-126 E	1.5926	50/50	1.5926
941331	AE2-129 C	2.4065	50/50	2.4065
941332	AE2-129 E	1.6043	50/50	1.6043
941351	AE2-131 C	2.4065	50/50	2.4065
941352	AE2-131 E	1.6043	50/50	1.6043
942121	AE2-224 C	7.8006	50/50	7.8006
942122	AE2-224 E	5.2004	50/50	5.2004
942351	AE2-248 C	3.5617	50/50	3.5617
942352	AE2-248 E	2.3745	50/50	2.3745
942491	AE2-262 C	14.8890	50/50	14.8890
942492	AE2-262 E	10.0054	50/50	10.0054
942501	AE2-263 C	13.9957	50/50	13.9957
942502	AE2-263 E	9.3443	50/50	9.3443
942511	AE2-264 C	15.7354	50/50	15.7354
942512	AE2-264 E	10.4902	50/50	10.4902
942961	AE2-316 C	6.0132	50/50	6.0132
942962	AE2-316 E	8.5748	50/50	8.5748
943751	AF1-043	21.0619	50/50	21.0619
944001	AF1-068 C O1	4.7786	50/50	4.7786
944002	AF1-068 E O1	2.6880	50/50	2.6880
944181	AF1-086 C O1	2.7086	50/50	2.7086
944182	AF1-086 E O1	11.7839	50/50	11.7839
944311	AF1-099 C	22.6015	50/50	22.6015
944312	AF1-099 E	15.0677	50/50	15.0677
944321	AF1-100 C	38.0220	50/50	38.0220
944322	AF1-100 E	25.3480	50/50	25.3480
944381	AF1-103 O1	1.8380	Adder	2.16
944471	AF1-112 C	4.7231	50/50	4.7231
944472	AF1-112 E	2.6567	50/50	2.6567
944671	AF1-132 C O1	4.4957	50/50	4.4957
944672	AF1-132 E O1	2.9971	50/50	2.9971
944691	AF1-134 C O1	2.0327	50/50	2.0327
944692	AF1-134 E O1	1.3551	50/50	1.3551
944751	AF1-140 C	1.0055	Adder	1.18
944752	AF1-140 E	0.6703	Adder	0.79
944771	AF1-142 C	36.1624	50/50	36.1624
944772	AF1-142 E	24.1083	50/50	24.1083
944841	AF1-149 C	4.3318	50/50	4.3318
944842	AF1-149 E	2.8878	50/50	2.8878

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
944881	AF1-153 C O1	2.2862	50/50	2.2862
944882	AF1-153 E O1	1.5242	50/50	1.5242
944901	AF1-155 C	2.2724	50/50	2.2724
944902	AF1-155 E	1.5150	50/50	1.5150
945121	AF1-177	0.4595	Adder	0.54
945161	AF1-181	0.2036	50/50	0.2036
945171	AF1-182	1.0204	50/50	1.0204
945181	AF1-183	0.2376	50/50	0.2376
945491	AF1-214 C	4.3304	50/50	4.3304
945492	AF1-214 E	2.8870	50/50	2.8870
945551	AF1-220 C	16.8514	50/50	16.8514
945552	AF1-220 E	11.2404	50/50	11.2404
946111	AF1-276 C	6.2673	Adder	7.37
946112	AF1-276 E	3.0869	Adder	3.63
946121	AF1-277 C	6.2673	Adder	7.37
946122	AF1-277 E	3.0869	Adder	3.63
946131	AF1-278 C	4.9952	Adder	5.88
946132	AF1-278 E	2.4845	Adder	2.92
946381	AF1-302 C	2.6258	50/50	2.6258
946382	AF1-302 E	3.5011	50/50	3.5011
946421	AF1-306 C	9.9270	50/50	9.9270
946422	AF1-306 E	39.7082	50/50	39.7082
WEC	WEC	0.3695	Confirmed LTF	0.3695
LGEE	LGEE	0.6407	Confirmed LTF	0.6407
CPL	CPL	0.3281	Confirmed LTF	0.3281
CBM-W2	CBM-W2	8.4930	Confirmed LTF	8.4930
TVA	TVA	1.3244	Confirmed LTF	1.3244
O-066	O-066	21.4906	Confirmed LTF	21.4906
CBM-S2	CBM-S2	3.6587	Confirmed LTF	3.6587
CBM-S1	CBM-S1	8.3411	Confirmed LTF	8.3411
G-007	G-007	3.3426	Confirmed LTF	3.3426
MEC	MEC	1.7829	Confirmed LTF	1.7829
CBM-W1	CBM-W1	15.4373	Confirmed LTF	15.4373

11.7.5 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41550311	200522	26SHADE GP	PENELEC	938380	AE1-071 TAP	PENELEC	1	PL_P42_100548	breaker	160.0	123.73	125.21	AC	2.4

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200812	26ALY HYDR	0.4762	50/50	0.4762
200852	26WARR RDG	0.1361	50/50	0.1361
235003	AC1-025 E	0.1804	50/50	0.1804
236828	01GRAYMONT	0.4640	50/50	0.4640
290086	Q-036 E	2.0495	Adder	2.41
293301	N-039 E	3.5676	Adder	4.2
294515	O38_P22	3.1216	Adder	3.67
919491	AA2-000	26.1120	Adder	30.72
930511	AB1-092	0.9588	Adder	1.13
936421	AD2-055	1.9739	Adder	2.32
936991	AD2-133 C	0.9223	Adder	1.09
936992	AD2-133 E	4.2184	Adder	4.96
939171	AE1-147 C	1.3841	50/50	1.3841
939172	AE1-147 E	0.9227	50/50	0.9227
940201	AE2-001 C	1.3886	50/50	1.3886
940202	AE2-001 E	0.9258	50/50	0.9258
940671	AE2-054 C	-0.5515	Adder	-0.65
940681	AE2-055 C	1.4580	50/50	1.4580
940682	AE2-055 E	0.9720	50/50	0.9720
941231	AE2-117 C	1.5364	50/50	1.5364
941232	AE2-117 E	1.0242	50/50	1.0242
941241	AE2-118 C	1.5364	50/50	1.5364
941242	AE2-118 E	1.0242	50/50	1.0242
941261	AE2-120 C	1.3910	50/50	1.3910
941262	AE2-120 E	0.9274	50/50	0.9274
941271	AE2-121 C	0.7317	50/50	0.7317
941272	AE2-121 E	0.4885	50/50	0.4885
941321	AE2-126 C	0.6617	Adder	0.78
941322	AE2-126 E	0.4411	Adder	0.52
941331	AE2-129 C	0.6652	Adder	0.78
941332	AE2-129 E	0.4435	Adder	0.52
941351	AE2-131 C	0.6652	Adder	0.78
941352	AE2-131 E	0.4435	Adder	0.52
942351	AE2-248 C	1.1433	50/50	1.1433
942352	AE2-248 E	0.7622	50/50	0.7622
942491	AE2-262 C	4.0855	Adder	4.81
942492	AE2-262 E	2.7455	Adder	3.23
942501	AE2-263 C	3.8404	Adder	4.52
942502	AE2-263 E	2.5641	Adder	3.02
942511	AE2-264 C	5.0510	50/50	5.0510
942512	AE2-264 E	3.3674	50/50	3.3674

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
943751	AF1-043	5.9217	Adder	6.97
944001	AF1-068 C O1	1.5337	50/50	1.5337
944002	AF1-068 E O1	0.8627	50/50	0.8627
944311	AF1-099 C	6.2018	Adder	7.3
944312	AF1-099 E	4.1346	Adder	4.86
944321	AF1-100 C	10.4754	Adder	12.32
944322	AF1-100 E	6.9836	Adder	8.22
944471	AF1-112 C	1.5163	50/50	1.5163
944472	AF1-112 E	0.8529	50/50	0.8529
944671	AF1-132 C O1	1.4428	50/50	1.4428
944672	AF1-132 E O1	0.9618	50/50	0.9618
944691	AF1-134 C O1	0.5643	Adder	0.66
944692	AF1-134 E O1	0.3762	Adder	0.44
944771	AF1-142 C	9.9229	Adder	11.67
944772	AF1-142 E	6.6153	Adder	7.78
944841	AF1-149 C	1.3914	50/50	1.3914
944842	AF1-149 E	0.9276	50/50	0.9276
944881	AF1-153 C O1	0.6331	Adder	0.74
944882	AF1-153 E O1	0.4221	Adder	0.5
944901	AF1-155 C	0.6293	Adder	0.74
944902	AF1-155 E	0.4196	Adder	0.49
945161	AF1-181	0.0561	Adder	0.07
945171	AF1-182	0.2810	Adder	0.33
945181	AF1-183	0.0655	Adder	0.08
945491	AF1-214 C	1.3909	50/50	1.3909
945492	AF1-214 E	0.9273	50/50	0.9273
945551	AF1-220 C	4.6797	Adder	5.51
945552	AF1-220 E	3.1215	Adder	3.67
946421	AF1-306 C	2.7481	Adder	3.23
946422	AF1-306 E	10.9925	Adder	12.93
WEC	WEC	0.0400	Confirmed LTF	0.0400
LGEE	LGEE	0.0593	Confirmed LTF	0.0593
CBM-W2	CBM-W2	0.4996	Confirmed LTF	0.4996
TVA	TVA	0.0238	Confirmed LTF	0.0238
O-066	O-066	1.5926	Confirmed LTF	1.5926
CHEOAH	CHEOAH	0.0110	Confirmed LTF	0.0110
CBM-S1	CBM-S1	0.2812	Confirmed LTF	0.2812
G-007	G-007	0.2829	Confirmed LTF	0.2829
MEC	MEC	0.1621	Confirmed LTF	0.1621
CALDERWOOD	CALDERWOOD	0.0094	Confirmed LTF	0.0094
CATAWBA	CATAWBA	0.0427	Confirmed LTF	0.0427
CBM-W1	CBM-W1	1.9391	Confirmed LTF	1.9391

11.7.6 Index 6

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41217821	235248	01SHINGL	AP	200513	26LEWISTWN	PENELEC	1	Base Case	single	491.0	117.33	118.51	AC	6.07

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200642	26SENECA#1	5.6020	80/20	5.6020
200643	26SENECA#2	5.2908	80/20	5.2908
200644	26SENECA#3	0.5272	80/20	0.5272
200649	26PENNTech	1.6014	80/20	1.6014
200665	26SHAWVL 3	9.7836	80/20	9.7836
200666	26SHAWVL 4	9.7549	80/20	9.7549
200715	26SHAWVL 1	6.2347	80/20	6.2347
200722	26SHAWVL 2	6.3940	80/20	6.3940
200887	26ARMNA MT	0.2802	80/20	0.2802
200898	26AA1-106	2.4075	80/20	2.4075
200905	26Q36	0.2522	80/20	0.2522
200913	26SHAW-D	0.2688	80/20	0.2688
200949	26X1-109	11.5046	80/20	11.5046
201201	26WRREN CT	1.1561	80/20	1.1561
201477	26Y2-055	3.3243	80/20	3.3243
203261	26BLOSSBCT	0.3250	80/20	0.3250
203283	26MANOR_T86	0.0181	80/20	0.0181
203349	26Z1-069 C	0.2175	80/20	0.2175
203351	26GROZ1-110	0.4264	80/20	0.4264
203352	26CANZ2-011	0.4264	80/20	0.4264
919201	AA1-144 OP	14.6276	80/20	14.6276
919491	AA2-000	117.4816	80/20	117.4816
922932	AB1-082 OP (Suspended)	2.0853	80/20	2.0853
923442	AB1-160 C	0.0621	80/20	0.0621
930511	AB1-092	4.3136	80/20	4.3136
934801	AD1-108	0.0321	80/20	0.0321
934811	AD1-109	0.0236	80/20	0.0236
936421	AD2-055	8.8809	80/20	8.8809
936991	AD2-133 C	2.5632	80/20	2.5632
939171	AE1-147 C	5.7857	80/20	5.7857
940201	AE2-001 C	5.8076	80/20	5.8076
940681	AE2-055 C	6.1405	80/20	6.1405
940861	AE2-074 C	1.5834	80/20	1.5834
941191	AE2-113 C	9.9578	80/20	9.9578
941261	AE2-120 C	5.8195	80/20	5.8195
941271	AE2-121 C	3.0559	80/20	3.0559
941321	AE2-126 C	2.7770	80/20	2.7770
941331	AE2-129 C	2.3051	80/20	2.3051
941351	AE2-131 C	2.3051	80/20	2.3051
941421	AE2-139 C	6.8573	80/20	6.8573
942351	AE2-248 C	4.8012	80/20	4.8012

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
942491	AE2-262 C	19.5050	80/20	19.5050
942501	AE2-263 C	18.3347	80/20	18.3347
942961	AE2-316 C	6.7968	80/20	6.7968
943751	AF1-043	26.6427	80/20	26.6427
944001	AF1-068 C O1	6.4463	80/20	6.4463
944181	AF1-086 C O1	2.3029	80/20	2.3029
944311	AF1-099 C	29.6086	80/20	29.6086
944321	AF1-100 C	48.0975	80/20	48.0975
944381	AF1-103 O1	2.2904	80/20	2.2904
944471	AF1-112 C	6.3629	80/20	6.3629
944671	AF1-132 C O1	6.0670	80/20	6.0670
944691	AF1-134 C O1	2.0707	80/20	2.0707
944771	AF1-142 C	47.3737	80/20	47.3737
944841	AF1-149 C	5.8208	80/20	5.8208
944881	AF1-153 C O1	2.7576	80/20	2.7576
944901	AF1-155 C	2.7377	80/20	2.7377
945121	AF1-177	0.5726	80/20	0.5726
945161	AF1-181	0.2569	80/20	0.2569
945171	AF1-182	1.2884	80/20	1.2884
945181	AF1-183	0.2885	80/20	0.2885
945331	AF1-198	0.2415	80/20	0.2415
945491	AF1-214 C	5.8188	80/20	5.8188
945551	AF1-220 C	19.7115	80/20	19.7115
946111	AF1-276 C	7.3318	80/20	7.3318
946121	AF1-277 C	7.3318	80/20	7.3318
946131	AF1-278 C	5.8436	80/20	5.8436
946381	AF1-302 C	2.9680	80/20	2.9680
946421	AF1-306 C	12.0982	80/20	12.0982
WEC	WEC	0.1616	Confirmed LTF	0.1616
LGEE	LGEE	0.2698	Confirmed LTF	0.2698
CPL	CPL	0.0318	Confirmed LTF	0.0318
CBM-W2	CBM-W2	3.4070	Confirmed LTF	3.4070
TVA	TVA	0.4914	Confirmed LTF	0.4914
CBM-S2	CBM-S2	0.8034	Confirmed LTF	0.8034
CBM-S1	CBM-S1	3.1865	Confirmed LTF	3.1865
MEC	MEC	0.7564	Confirmed LTF	0.7564
CBM-W1	CBM-W1	7.0431	Confirmed LTF	7.0431

11.7.7 Index 7

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41806793	938380	AE1-071 TAP	PENELEC	200520	26ROXBURY	PENELEC	1	PL_P22_100582	bus	160.0	163.52	165.0	AC	2.4

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
200812	26ALY HYDR	0.4759	50/50	0.4759
200852	26WARR RDG	0.1360	50/50	0.1360
235003	AC1-025 E	0.1803	50/50	0.1803
236828	01GRAYMONT	0.4637	50/50	0.4637
290086	Q-036 E	2.0461	Adder	2.41
293301	N-039 E	3.5621	Adder	4.19
294515	O38_P22	3.1168	Adder	3.67
919491	AA2-000	26.0766	Adder	30.68
930511	AB1-092	0.9575	Adder	1.13
936421	AD2-055	1.9712	Adder	2.32
936991	AD2-133 C	0.9208	Adder	1.08
936992	AD2-133 E	4.2114	Adder	4.95
938381	AE1-071 C	41.3033	50/50	41.3033
938382	AE1-071 E	25.2742	50/50	25.2742
939171	AE1-147 C	1.3831	50/50	1.3831
939172	AE1-147 E	0.9221	50/50	0.9221
940201	AE2-001 C	1.3877	50/50	1.3877
940202	AE2-001 E	0.9251	50/50	0.9251
940671	AE2-054 C	-0.5507	Adder	-0.65
940681	AE2-055 C	1.4572	50/50	1.4572
940682	AE2-055 E	0.9714	50/50	0.9714
941231	AE2-117 C	1.5355	50/50	1.5355
941232	AE2-117 E	1.0237	50/50	1.0237
941241	AE2-118 C	1.5355	50/50	1.5355
941242	AE2-118 E	1.0237	50/50	1.0237
941261	AE2-120 C	1.3902	50/50	1.3902
941262	AE2-120 E	0.9268	50/50	0.9268
941271	AE2-121 C	0.7311	50/50	0.7311
941272	AE2-121 E	0.4882	50/50	0.4882
941321	AE2-126 C	0.6608	Adder	0.78
941322	AE2-126 E	0.4405	Adder	0.52
941331	AE2-129 C	0.6642	Adder	0.78
941332	AE2-129 E	0.4428	Adder	0.52
941351	AE2-131 C	0.6642	Adder	0.78
941352	AE2-131 E	0.4428	Adder	0.52
942351	AE2-248 C	1.1426	50/50	1.1426
942352	AE2-248 E	0.7617	50/50	0.7617
942491	AE2-262 C	4.0821	Adder	4.8
942492	AE2-262 E	2.7432	Adder	3.23
942501	AE2-263 C	3.8372	Adder	4.51
942502	AE2-263 E	2.5619	Adder	3.01

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
942511	AE2-264 C	5.0467	50/50	5.0467
942512	AE2-264 E	3.3645	50/50	3.3645
943751	AF1-043	5.9137	Adder	6.96
944001	AF1-068 C O1	1.5327	50/50	1.5327
944002	AF1-068 E O1	0.8621	50/50	0.8621
944311	AF1-099 C	6.1967	Adder	7.29
944312	AF1-099 E	4.1311	Adder	4.86
944321	AF1-100 C	10.4639	Adder	12.31
944322	AF1-100 E	6.9760	Adder	8.21
944471	AF1-112 C	1.5153	50/50	1.5153
944472	AF1-112 E	0.8523	50/50	0.8523
944671	AF1-132 C O1	1.4418	50/50	1.4418
944672	AF1-132 E O1	0.9612	50/50	0.9612
944691	AF1-134 C O1	0.5632	Adder	0.66
944692	AF1-134 E O1	0.3755	Adder	0.44
944771	AF1-142 C	9.9147	Adder	11.66
944772	AF1-142 E	6.6098	Adder	7.78
944841	AF1-149 C	1.3904	50/50	1.3904
944842	AF1-149 E	0.9270	50/50	0.9270
944881	AF1-153 C O1	0.6322	Adder	0.74
944882	AF1-153 E O1	0.4215	Adder	0.5
944901	AF1-155 C	0.6284	Adder	0.74
944902	AF1-155 E	0.4189	Adder	0.49
945161	AF1-181	0.0560	Adder	0.07
945171	AF1-182	0.2807	Adder	0.33
945181	AF1-183	0.0654	Adder	0.08
945491	AF1-214 C	1.3901	50/50	1.3901
945492	AF1-214 E	0.9267	50/50	0.9267
945551	AF1-220 C	4.6719	Adder	5.5
945552	AF1-220 E	3.1163	Adder	3.67
946421	AF1-306 C	2.7444	Adder	3.23
946422	AF1-306 E	10.9774	Adder	12.91
WEC	WEC	0.0391	Confirmed LTF	0.0391
LGEE	LGEE	0.0576	Confirmed LTF	0.0576
CBM-W2	CBM-W2	0.4832	Confirmed LTF	0.4832
TVA	TVA	0.0210	Confirmed LTF	0.0210
O-066	O-066	1.5994	Confirmed LTF	1.5994
CHEOAH	CHEOAH	0.0120	Confirmed LTF	0.0120
CBM-S1	CBM-S1	0.2641	Confirmed LTF	0.2641
G-007	G-007	0.2829	Confirmed LTF	0.2829
MEC	MEC	0.1573	Confirmed LTF	0.1573
CALDERWOOD	CALDERWOOD	0.0104	Confirmed LTF	0.0104
CATAWBA	CATAWBA	0.0427	Confirmed LTF	0.0427
CBM-W1	CBM-W1	1.8890	Confirmed LTF	1.8890

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA1-106	Grover II 34.5kV	Withdrawn
AA1-144	East Towanda-Grover 230kV	Engineering and Procurement
AA2-000	N/A	N/A
AB1-082	Potter 46kV	Suspended
AB1-092	Moshannon-East Towanda 230kV	Active
AB1-160	Gold-Sabinsville 115kV	In Service
AC1-025	Dale Summit	In Service
AD1-108	Grover 34 kV	In Service
AD1-109	Canton 34.5 kV	In Service
AD2-055	Moshannon-East Towanda 230 kV	Active
AD2-133	Eagle Valley 115kV	Active
AE1-071	Shade Gap-Roxbury 115 kV	Active
AE1-128	Bedford North-Wills Mounain 115 kV	Active
AE1-147	Bellefonte 46 kV	Engineering and Procurement
AE2-001	Nittany-Zion 46 kV	Active
AE2-054	Warfordsburg 34 kV	Engineering and Procurement
AE2-055	Shingletown-Boalsburg 46 kV	Engineering and Procurement
AE2-074	Potter 46 kV	Active
AE2-113	Farmers Valley-Ridgeway 115 kV	Active
AE2-117	ABW Tap-Alexandria 46 kV	Active
AE2-118	ABW Tap-Williamsburg 46 kV	Active
AE2-120	Graymont-Zion 46 kV	Active
AE2-121	Milesburg-Tanney Junction 46 kV	Active
AE2-126	Dubois-Curwensville 34.5 kV	Active
AE2-129	Philipsburg-Clarence 34.5 kV	Active
AE2-131	Philipsburg-Karthus 34.5	Active
AE2-139	East Towanda-Grover 230 kV	Active
AE2-224	Bearrock-Johnstown 230 kV	Active
AE2-248	Fillmore-Thompson Farm 46 kV	Active
AE2-262	Moshannon-Milesburg 230 kV	Active
AE2-263	Moshannon-Milesburg 230 kV	Active
AE2-264	Altoona-Raystown 230 kV	Active
AE2-316	Brookville-Squab Hollow 138 kV	Active
AF1-043	Moshannon-East Towanda 230 kV	Active
AF1-068	Boalsburg-Centre Hall 46 kV	Active
AF1-086	Madera-Westover South 115 kV	Active
AF1-099	Moshannon-Milesburg 230 kV	Active
AF1-100	Shawville-Moshannon 230 kV	Active
AF1-103	Warren 34.5 kV	Active
AF1-112	Centre Hall-Boalsburg 46 kV	Active
AF1-132	Shingletown-Boalsburg 46 kV	Active
AF1-134	Philipsburg-Madera 34.5 kV	Active
AF1-140	Claysburg 23 kV	Engineering and Procurement

Queue Number	Project Name	Status
AF1-142	Moshannon-Milesburg 230 kV	Active
AF1-149	Graymont-Zion 46 kV	Active
AF1-153	Motion-Ridgeway 46 kV	Active
AF1-155	Paper City-Wilcox 46 kV	Engineering and Procurement
AF1-177	Warren 115 kV	Partially in Service - Under Construction
AF1-181	Shawville 3 230 kV	Partially in Service - Under Construction
AF1-182	Shawville 4 230 kV	Partially in Service - Under Construction
AF1-183	Shawville 1 230 kV	Partially in Service - Under Construction
AF1-198	Blossburg #1 CT 34.5 kV	Partially in Service - Under Construction
AF1-214	Nittany-Zion 46 kV	Active
AF1-220	Ridgway-Whetstone 115 kV	Active
AF1-276	Lewis Run-Pierce Brook 230 kV	Active
AF1-277	Lewis Run-Pierce Brook 2 230 kV	Active
AF1-278	Lewis Run-Pierce Brook 3 230 kV	Active
AF1-302	Brookville-Squab Hollow 138 kV	Active
AF1-306	Squab Hollow 230 kV	Active
V3-030	St. Benedict-Patton 46kV	In Service
X1-109	E. Towanda 230kV	In Service
Y2-055	Elm Street 34.5kV	Withdrawn
Z1-069	Gold-Sabinsville 115kV	In Service
Z1-110	Grover 34kV	In Service
Z2-011	Canton 34.5kV	In Service

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
PN-P2-4-PN-230-14GT	CONTINGENCY 'PN-P2-4-PN-230-14GT' /* LEWISTOWN STUCK 230KV BUS TIE BREAKER DISCONNECT BRANCH FROM BUS 200513 TO BUS 235248 CKT 1 /* 26LEWISTWN 230 01SHINGL 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200517 CKT 1 /* 26LEWISTWN 230 26RAYSTOWN 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200512 TO BUS 200548 CKT 1/* 26LEWISTWN 230 26LEWISTWN 115 26LEWISTWN 46.00 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200512 TO BUS 200548 CKT 2/* 26LEWISTWN 230 26LEWISTWN 115 26LEWISTWN 46.00 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200512 CKT 3 /* 26LEWISTWN 230 26LEWISTWN 115 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200531 CKT 1 /* 26LEWISTWN 230 26YEAGRTWN 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200531 CKT 2 /* 26LEWISTWN 230 26YEAGRTWN 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 208005 CKT 1 /* 26LEWISTWN 230 JUNI BU2 230 REDUCE BUS 200513 SHUNT BY 100 PERCENT /* 26LEWISTWN 230 END
PN-P1-2-PN-115-094-A	CONTINGENCY 'PN-P1-2-PN-115-094-A' /* ROXBURY - SHADE GAP 115KV DISCONNECT BRANCH FROM BUS 200520 TO BUS 938380 CKT 1 /* 26ROXBURY 115 AE1-071 TAP 115 END
PN-P2-2-PN-115-032	CONTINGENCY 'PN-P2-2-PN-115-032' /* ROXBURY 115KV BUS FAULT DISCONNECT BUS 200520 /* 26ROXBURY 115 END
PL_P42_100548	CONTINGENCY 'PL_P42_100548' /* AT JUNIATA SUB 230KV BUS TIE CB FAILED" DISCONNECT BUS 208005 /* DISCONNECT BUS 208004 /* END
AP-P2-3-WP-230-446T	CONTINGENCY 'AP-P2-3-WP-230-446T' /* ELKO-MOSHANNON STK BKR AT ELKO DISCONNECT BRANCH FROM BUS 200726 TO BUS 235175 CKT 1 /* 26SHAWVL 2 230 01ELKO 230 DISCONNECT BRANCH FROM BUS 235158 TO BUS 235175 CKT 1 /* 01CARB 230 01ELKO 230 DISCONNECT BRANCH FROM BUS 235175 TO BUS 235236 CKT 1 /* 01ELKO 230 01QUEHAN 230 DISCONNECT BRANCH FROM BUS 235220 TO BUS 235236 CKT 1 /* 01MOSHAN 230 01QUEHAN 230 DISCONNECT BRANCH FROM BUS 235236 TO BUS 236732 CKT 81 /* 01QUEHAN 230 01QUEHANNA 46 END

Contingency Name	Contingency Definition
PN-P1-2-PN-230-012A-A	CONTINGENCY 'PN-P1-2-PN-230-012A-A' /* QUEUE AA1-144 - MARSHALL 230KV DISCONNECT BRANCH FROM BUS 200857 TO BUS 200701 CKT 1 /* 26MARSHALL 230 26GROVER 230 DISCONNECT BRANCH FROM BUS 200701 TO BUS 941420 CKT 1 /* 26GROVER 230 AE2- 139 TAP 230 DISCONNECT BRANCH FROM BUS 200701 TO BUS 200702 CKT 1 /* 26GROVER 230 26GROVER 35 END
AP-P2-3-WP-230-460T	CONTINGENCY 'AP-P2-3-WP-230-460T' /* 467 DISCONNECT BRANCH FROM BUS 200513 TO BUS 235248 CKT 1 /* 26LEWISTWN 230 01SHINGL 230 DISCONNECT BRANCH FROM BUS 200726 TO BUS 235248 CKT 1 /* 26SHAWVL 2 230 01SHINGL 230 DISCONNECT BRANCH FROM BUS 235248 TO BUS 235970 CKT 1 /* 01SHINGL 230 01DALE 230 DISCONNECT BRANCH FROM BUS 236711 TO BUS 235248 CKT 81 /* 01SHINGLTN 46 01SHINGL 230 DISCONNECT BRANCH FROM BUS 236711 TO BUS 235248 CKT 82 /* 01SHINGLTN 46 01SHINGL 230 END
AP-P1-2-WP-230-323T	CONTINGENCY 'AP-P1-2-WP-230-323T' /* SHINGLETOWN-LEWISTOWN 230KV APS-PN TIE DISCONNECT BRANCH FROM BUS 235248 TO BUS 200513 CKT 1 /* 01SHINGL 230 26LEWISTWN 230 END
PN-P2-2-PN-230-007T	CONTINGENCY 'PN-P2-2-PN-230-007T' /* LEWISTOWN #2 230KV BUS DISCONNECT BRANCH FROM BUS 200513 TO BUS 200531 CKT 2 /* 26LEWISTWN 230 26YEAGRTWN 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 235248 CKT 1 /* 26LEWISTWN 230 01SHINGL 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200517 CKT 1 /* 26LEWISTWN 230 26RAYSTOWN 230 END
PN-P2-3-PN-230-14CT	CONTINGENCY 'PN-P2-3-PN-230-14CT' /* LEWISTOWN STUCK 230KV BREAKER - SHINGLETOWN DISCONNECT BRANCH FROM BUS 200513 TO BUS 200517 CKT 1 /* 26LEWISTWN 230 26RAYSTOWN 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 235248 CKT 1 /* 26LEWISTWN 230 01SHINGL 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200531 CKT 2 /* 26LEWISTWN 230 26YEAGRTWN 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200512 TO BUS 200548 CKT 2/* 26LEWISTWN 230 26LEWISTWN 115 26LEWISTWN 46.00 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200512 CKT 3 /* 26LEWISTWN 230 26LEWISTWN 115 END

Contingency Name	Contingency Definition
PN-P2-3-PN-230-14BT	CONTINGENCY 'PN-P2-3-PN-230-14BT' /* LEWISTOWN STUCK 230KV BREAKER B5 (RAYSTOWN) DISCONNECT BRANCH FROM BUS 200513 TO BUS 200517 CKT 1 /* 26LEWISTWN 230 26RAYSTOWN 230 DISCONNECT BRANCH FROM BUS 200517 TO BUS 200539 CKT 1 /* 26RAYSTOWN 230 26RAYSTOWN 46 DISCONNECT BRANCH FROM BUS 200513 TO BUS 235248 CKT 1 /* 26LEWISTWN 230 01SHINGL 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200531 CKT 2 /* 26LEWISTWN 230 26YEAGRTWN 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200512 TO BUS 200548 CKT 1/* 26LEWISTWN 230 26LEWISTWN 115 26LEWISTWN 46.00 END
Base Case	
PL_P12_100618	CONTINGENCY 'PL_P12_100618' /* JUNIATA 230/69KV TR4 OUT" DISCONNECT BRANCH FROM BUS 208005 TO BUS 209997 CKT 4 /* DISCONNECT BRANCH FROM BUS 208005 TO BUS 207955 CKT 1 /* DISCONNECT BRANCH FROM BUS 208005 TO BUS 200009 CKT 2 /* DISCONNECT BRANCH FROM BUS 208005 TO BUS 208006 CKT 1 /* DISCONNECT BRANCH FROM BUS 208005 TO BUS 208004 CKT 1 /* DISCONNECT BRANCH FROM BUS 208005 TO BUS 200513 CKT 1 /* END
PN-P2-3-PN-115-55A	CONTINGENCY 'PN-P2-3-PN-115-55A' /* 683 DISCONNECT BRANCH FROM BUS 200520 TO BUS 938380 CKT 1 /* 26ROXBURY 115 AE1-071 TAP 115 DISCONNECT BRANCH FROM BUS 200860 TO BUS 200520 CKT 3 /* 26ROXB SB2 23 26ROXBURY 115 DISCONNECT BRANCH FROM BUS 200504 TO BUS 200520 CKT 1 /* 26CARLISLE 115 26ROXBURY 115 DISCONNECT BRANCH FROM BUS 200504 TO BUS 202803 CKT 2 /* 26CARLISLE 115 26CARL PK2 23 DISCONNECT BRANCH FROM BUS 200532 TO BUS 200520 CKT 2 /* 26ROXBURY 138 26ROXBURY 115 DISCONNECT BRANCH FROM BUS 200520 TO BUS 200554 CKT 1 /* 26ROXBURY 115 26ROXB SB1 23 END

Contingency Name	Contingency Definition
PN-P2-3-PN-230-14FT	CONTINGENCY 'PN-P2-3-PN-230-14FT' /* LEWISTOWN STUCK 230KV BREAKER B34 (YEAGERTWON #2) DISCONNECT BRANCH FROM BUS 200513 TO BUS 200517 CKT 1 /* 26LEWISTWN 230 26RAYSTOWN 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 235248 CKT 1 /* 26LEWISTWN 230 01SHINGL 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200531 CKT 2 /* 26LEWISTWN 230 26YEAGRTWN 230 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200512 TO BUS 200548 CKT 2/* 26LEWISTWN 230 26LEWISTWN 115 26LEWISTWN 46.00 DISCONNECT BRANCH FROM BUS 200531 TO BUS 200550 CKT 2 /* 26YEAGRTWN 230 26YEAGRTWN 46 DISCONNECT BRANCH FROM BUS 200513 TO BUS 200512 CKT 3 /* 26LEWISTWN 230 26LEWISTWN 115 END
AP-P2-2-WP-230-001T	CONTINGENCY 'AP-P2-2-WP-230-001T' DISCONNECT BRANCH FROM BUS 235175 TO BUS 235158 CKT 1 DISCONNECT BRANCH FROM BUS 235175 TO BUS 235236 CKT 1 DISCONNECT BRANCH FROM BUS 235175 TO BUS 200726 CKT 1 END
AP-P1-3-WP-230-326T	CONTINGENCY 'AP-P1-3-WP-230-326T' /* SHINGLETOWN #82 230/46KV XFMR DISCONNECT BRANCH FROM BUS 235248 TO BUS 236711 CKT 82 /* 01SHINGL 230 01SHINGLTN 46 DISCONNECT BRANCH FROM BUS 235248 TO BUS 200513 CKT 1 /* 01SHINGL 230 26LEWISTWN 230 DISCONNECT BRANCH FROM BUS 235248 TO BUS 235970 CKT 1 /* 01SHINGL 230 01DALE 230 DISCONNECT BUS 237058 /*SHINGLETOWN 230 KV CAPACITOR END
PL_P22_100582	CONTINGENCY 'PL_P22_100582' /* JUNIATA 230KV BUS 2 DISCONNECT BUS 208005 /* END

12 Light Load Analysis

Light Load Analysis is not performed for solar projects.

13 Short Circuit Analysis

The following Breakers are overdutied:

None

14 Stability and Reactive Power

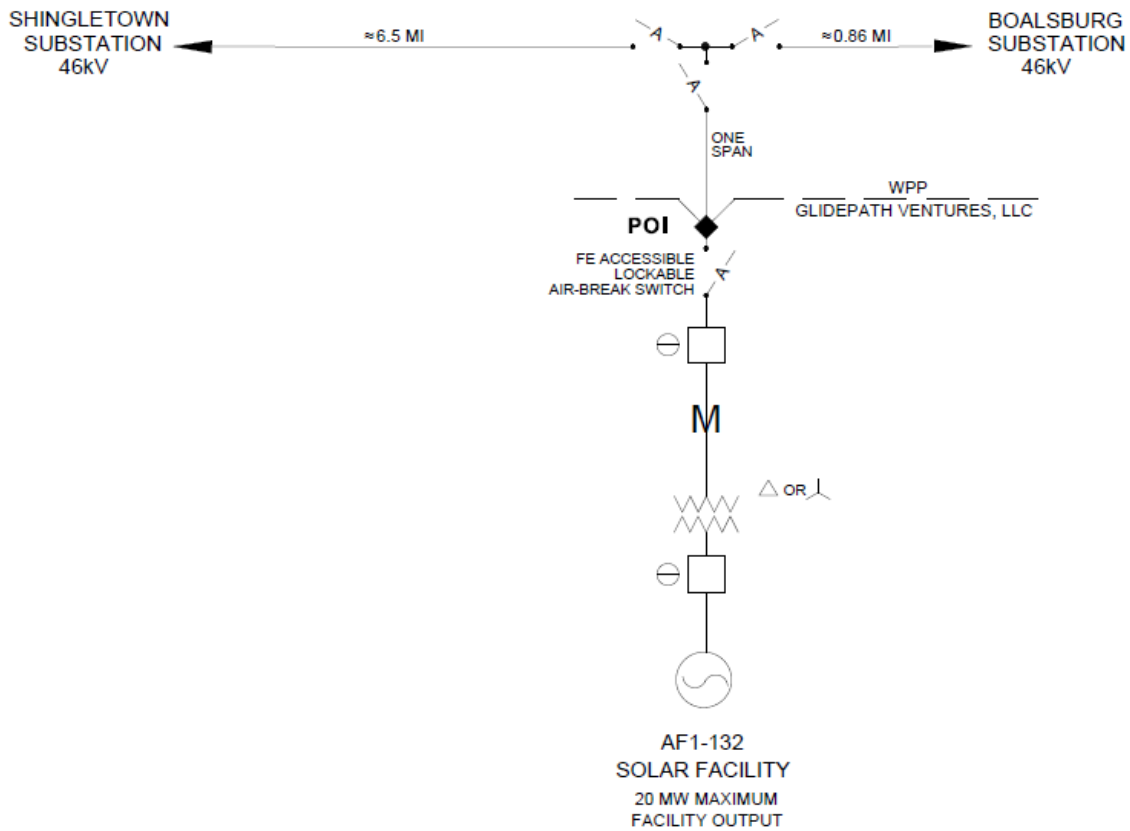
(Summary of the VAR requirements based upon the results of the dynamic studies)

To be determined in the Facilities Study Phase.

15 Affected Systems

None

16 Attachment 1: One Line Diagram



◆ = POI (POINT OF INTERCONNECTION) LOCATED AT INTERCONNECTION CUSTOMER'S DEAD-END STRUCTURE WHERE WEST PENN POWER'S (WPP) OWNED 46kV TRANSMISSION LINE TERMINATES

M = REVENUE METERING FOR INTERCONNECTION CUSTOMER IS OWNED, OPERATED, AND MAINTAINED BY WEST PENN POWER.

⊖ = SCADA INDICATION